

Analysis of arthropod communities in Hawaii using METE Theory

Setup

```
library(meteR)  
library(ggplot2)  
library(dplyr)
```

Attaching package: 'dplyr'

The following objects are masked from 'package:stats':

filter, lag

The following objects are masked from 'package:base':

intersect, setdiff, setequal, union

```
library(tidyr)  
library(vegan)
```

Loading required package: permute

```
library(metafor)
```

Loading required package: Matrix

Attaching package: 'Matrix'

The following objects are masked from 'package:tidyr':

expand, pack, unpack

Loading required package: metadat

Loading required package: numDeriv

Loading the 'metafor' package (version 5.0-1). For an introduction to the package please type: help(metafor)

Attaching package: 'metafor'

The following object is masked from 'package:vegan':

permutest

```
library(HawaiiArthMETE)
```

Attaching package: 'HawaiiArthMETE'

The following object is masked from 'package:meteR':

arth

```
# load data  
data(arth)
```

```
e <- meteESF(1:10, rpois(10, 3) + 1)  
foo <- logLikZ(sad(e))
```

```
foo
```

```
$z
```

```
'log Lik.' 0.4239621 (df=2)
```

```
$sim
```

```
NULL
```

METE calculations

```
arth_mete <- split(arth, arth[, c("site", "tree")], drop = TRUE) |>
  lapply(function(x) {
    # the esf for this grouping
    esf <- meteESF(spp = x$species_code,
                  abund = x$abundance,
                  power = x$metabolic_rate)

    # sad and ipd
    s <- sad(esf)
    p <- ipd(esf)

    data.frame(
      # sample info
      site = x$site[1],
      site_name = x$site_name[1],
      tree = x$tree[1],
      flow_age_my = x$flow_age_my[1],

      # z2 values
      z2_sad = logLikZ(s)$z,
      z2_ipd = logLikZ(p)$z,

      # prop non-native for abundance and power
      prop_n_non_nat = sum(
        x$abundance[!(x$origin %in% c("endemic", "indigenous"))]
      ) /
        sum(x$abundance),
      prop_p_non_nat = sum(
        x$metabolic_rate[!(x$origin %in% c("endemic", "indigenous"))]
      ) /
```

```
sum(x$metabolic_rate),

# differences in extreme values
n1_diff = mean(meteDist2Rank(s) == 1) - mean(s$data == 1),
nmax_diff = sum(meteDist2Rank(s)[1:3]) - sum(s$data[1:3]),
p1_diff = mean(meteDist2Rank(p) <= 2) - mean(p$data <= 2),
pmax_diff = sum(meteDist2Rank(p)[1:3]) - sum(p$data[1:3])
)
```

[illegible]


```
arth_mete <- do.call(rbind, arth_mete)
```

```
outliers <- filter(arth_mete, nmax_diff < -300)
```

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sites. To further understand these two samples and determine if they should remain in the analyses we look at their composition. Table 1 shows the three most abundant species for the Laupāhoehoe outlier sample while Table 2 shows the three most abundant species for the ‘r Kohala outlier sample.

```
filter(arth, tree == outliers$tree[1] & site == outliers$site[1]) |>
  summarize(n = n(), .by = c(genus, species, origin)) |>
  arrange(desc(n)) |>
  head(3) |>
  knitr::kable(
    caption = paste(
      "Top three most abundant taxa from",
      levels(arth$site_name)[as.numeric(outliers$site[1])],
      "outlier sample"
    )
  )
```

Table 1: Top three most abundant taxa from Laupāhoehoe outlier sample

genus	species	origin	n
Greenidea	formosana	adventive	423
Salina	celebensis	adventive	330
Entomobrya	socia	adventive	69

```
filter(arth, tree == outliers$tree[2] & site == outliers$site[2]) |>
  summarize(n = n(), .by = c(genus, species, origin)) |>
  arrange(desc(n)) |>
  head(3) |>
  knitr::kable(
    caption = paste(
      "Top three most abundant taxa from",
      levels(arth$site_name)[as.numeric(outliers$site[2])],
      "outlier sample"
    )
  )
```

Table 2: Top three most abundant taxa from Kohala outlier sample

genus	species	origin	n
Salina	celebensis	adventive	410

Table 2: Top three most abundant taxa from Kohala outlier sample

genus	species	origin	n
Forcipomyia	hardyi	endemic	92
Kilauella	sp_3	endemic	46

In both outliers the most abundant species is non-native, and in the case of Laupāhoehoe, all top three most abundant species are non-native. Laupāhoehoe and Kohala share *Salina celebensis* in common which is a Collembola known to reach high densities in Hawai'i [Christiansen 1992]. The abundances of *Salina celebensis* and *Greenidea formosana* in these samples are significant outliers compared to the maximum abundance found in any other sample as shown in Figure 1.

```
crazy_abund <- filter(arth,
                      (site %in% outliers$site & tree %in% outliers$tree)) |>
  summarize(n = n(), .by = c(site, tree, species_code)) |>
  arrange(desc(n)) |>
  head(3)

summarize(arth, .by = c(site, tree),
          nmax = max(table(species_code))) |>
  filter(nmax < 300) |>
  ggplot(aes(nmax)) +
  geom_histogram(bins = 20) +
  geom_segment(data = crazy_abund,
              mapping = aes(x = n, y = rep(10, 3),
                           xend = n, yend = rep(0, 3)),
              color = "red",
              arrow = arrow(length = unit(1, "char")),
              inherit.aes = FALSE) +
  xlab("Maximum abundances") +
  ylab("Count") +
  this_ms_look()
```

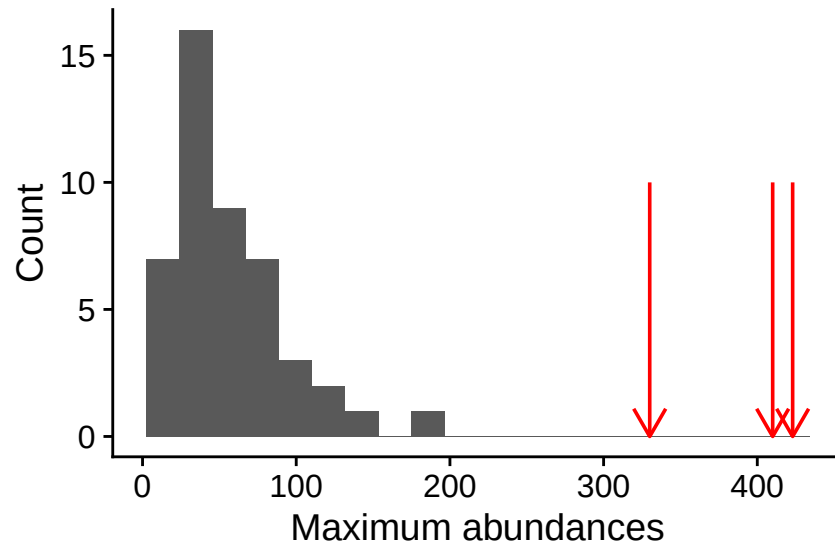


Figure 1

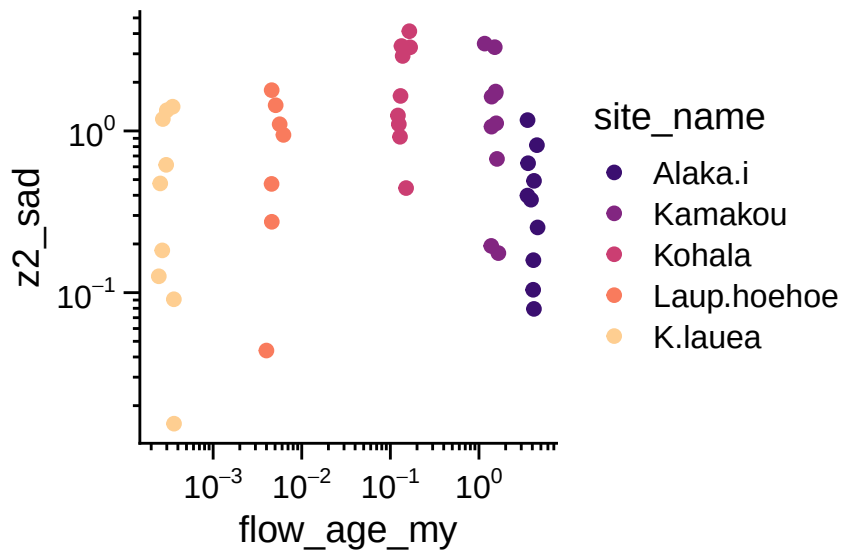
Because these samples are so unusual we decided to drop them from all further analyses presented in the main text. However, we retain the outlier samples in the supplement to validate that our results are not biased by their removal.

```
# remove from mete results
arth_mete_out <- arth_mete
arth_mete <- filter(arth_mete,
                    !((site %in% outliers$site) & (tree %in% outliers$tree)))

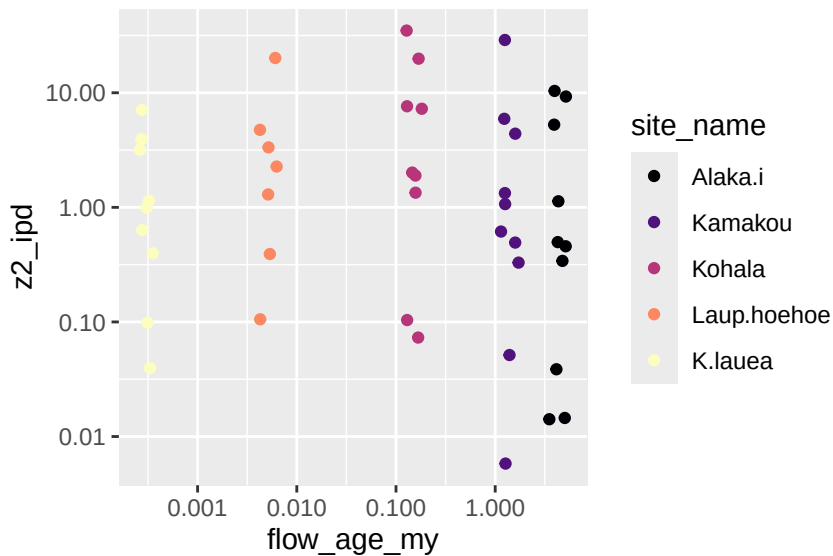
# remove from arth data frame
arth_out <- arth
arth <- filter(arth,
               !((site %in% outliers$site) & (tree %in% outliers$tree)))
```

Deviations from METE across the chronosequence

```
ggplot(arth_mete, aes(flow_age_my, z2_sad, color = site_name)) +
  geom_jitter(width = 0.1) +
  this_ms_look(log_x = TRUE, log_y = TRUE)
```



```
ggplot(arth_mete, aes(flow_age_my, z2_ipd, color = site_name)) +
  geom_jitter(width = 0.1) +
  scale_x_log10() +
  scale_y_log10() +
  scale_color_viridis_d(option = "magma")
```

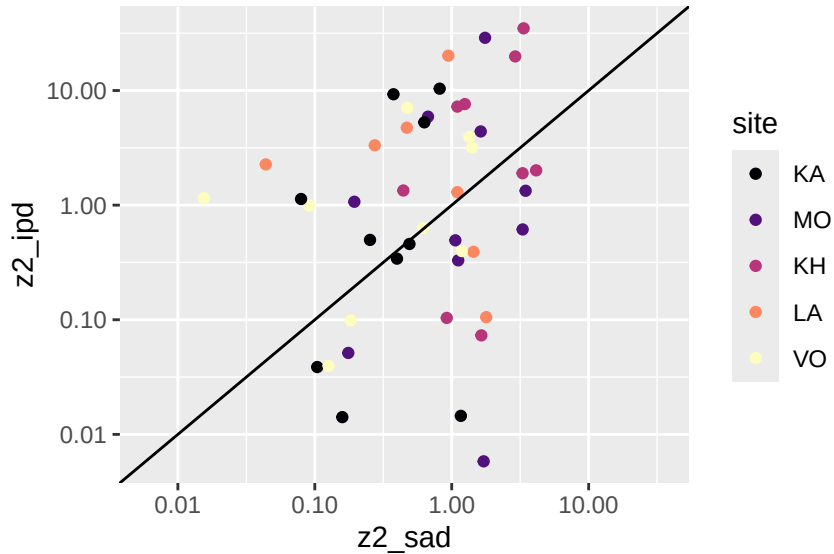


How do observed patterns deviate from METE?

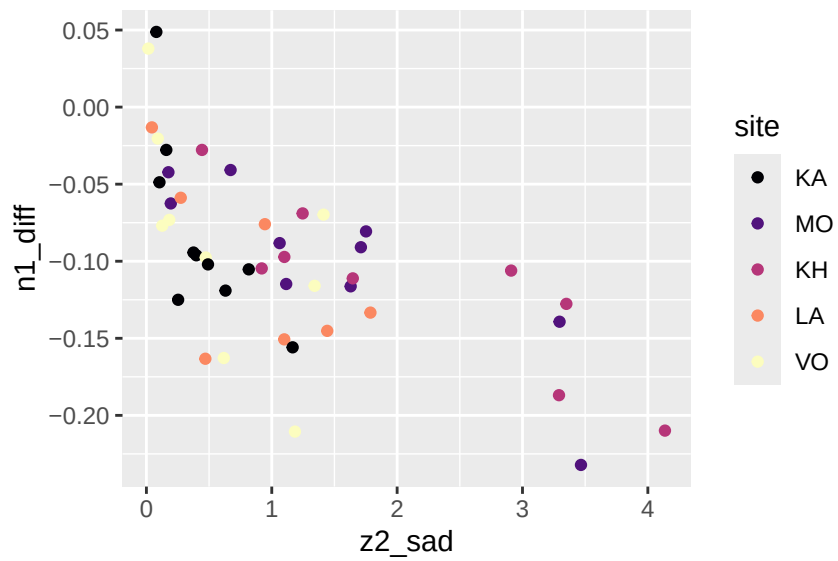
?@fig-sad-how shows that when METE predictions deviate from observed SADs, METE under-predicts both rarity and dominance compared to observed SADs. This means that

observed SADs are more uneven than the METE prediction.

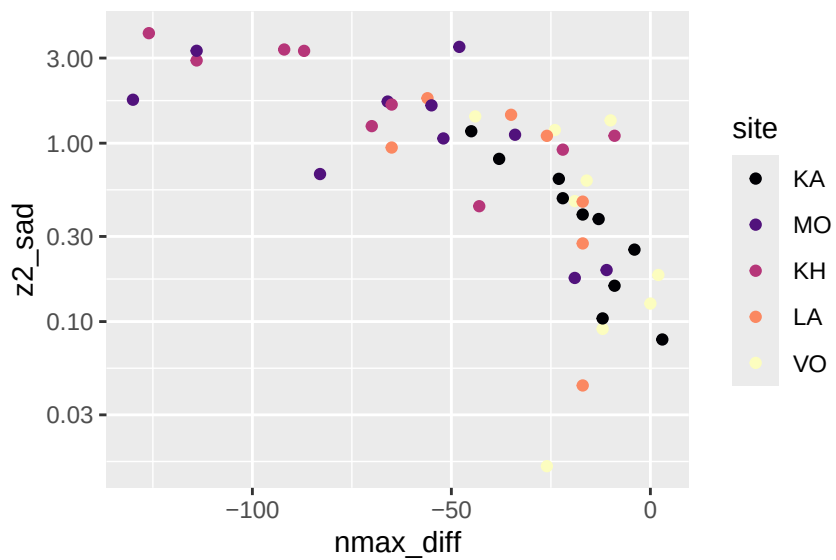
```
ggplot(arth_mete, aes(z2_sad, z2_ipd, color = site)) +  
  geom_point() +  
  scale_x_log10(limits = range(select(arth_mete, starts_with("z2")))) +  
  scale_y_log10(limits = range(select(arth_mete, starts_with("z2")))) +  
  scale_color_viridis_d(option = "magma") +  
  geom_abline(slope = 1, intercept = 0)
```



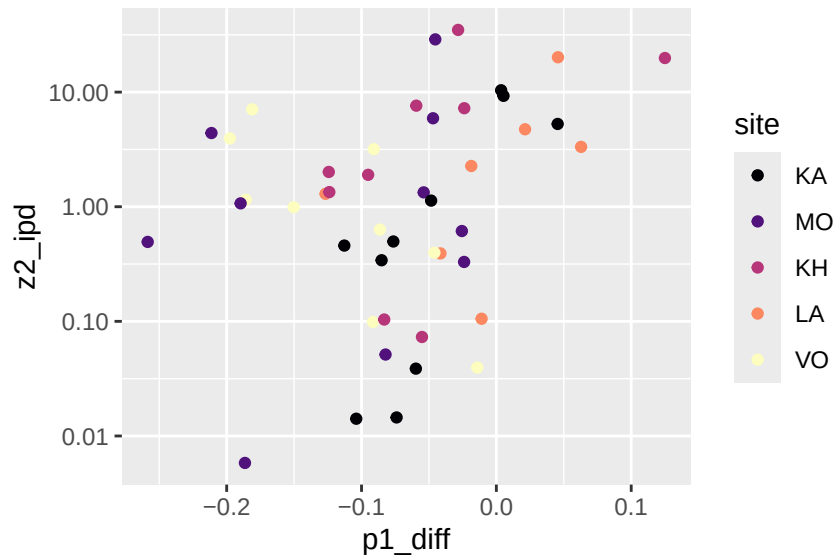
```
ggplot(arth_mete, aes(z2_sad, n1_diff, color = site)) +  
  geom_point() +  
  scale_color_viridis_d(option = "magma")
```



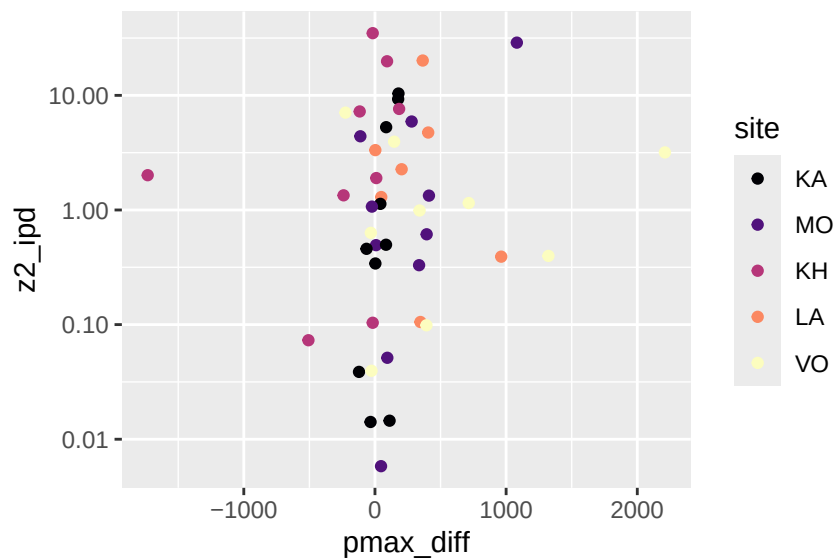
```
ggplot(arth_mete, aes(nmax_diff, z2_sad, color = site)) +
  geom_point() +
  scale_y_log10() +
  scale_color_viridis_d(option = "magma")
```



```
ggplot(arth_mete, aes(p1_diff, z2_ipd, color = site)) +
  geom_point() +
  scale_y_log10() +
  scale_color_viridis_d(option = "magma")
```



```
ggplot(arth_mete, aes(pmax_diff, z2_ipd, color = site)) +
  geom_point() +
  scale_y_log10() +
  scale_color_viridis_d(option = "magma")
```



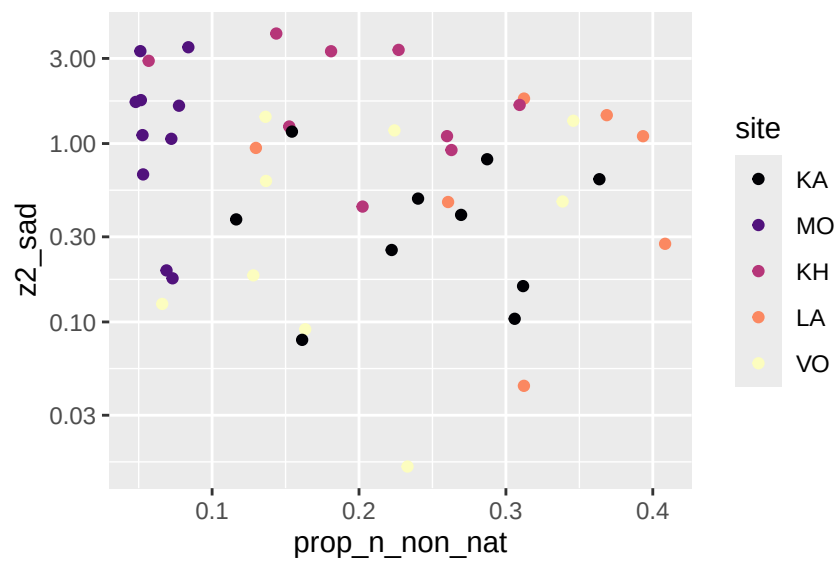
How do METE and the real data differ

Which one over-predicts rarity?

Comparing deviations from METE with potential explanatory variables

Deviation vs. proportion that are invasive

```
ggplot(arth_mete, aes(prop_n_non_nat, z2_sad, color = site)) +  
  geom_point() +  
  scale_y_log10() +  
  scale_color_viridis_d(option = "magma")  
  
ggplot(arth_mete, aes(prop_p_non_nat, z2_ipd, color = site)) +  
  geom_point() +  
  scale_y_log10() +  
  scale_color_viridis_d(option = "magma")
```



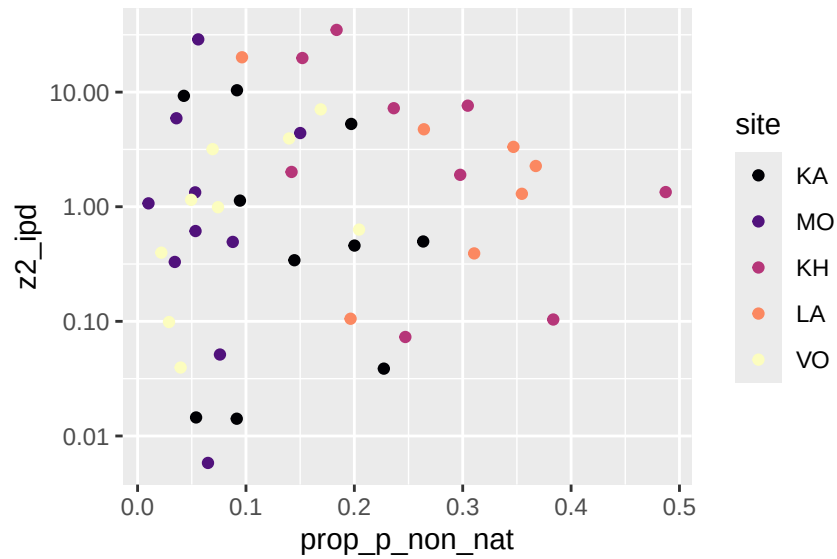


Figure 3

Deviation vs. dissimilarity

```
z_diss <- function(x, B = 500) {
  obs_dist <- x |> vegdist() |> mean()

  sim <- nullmodel(x, "quasiswap_count") |>
    simulate(nsim = B)

  sim_mean_dist <- sapply(1:dim(sim)[3], function(i) {
    vegdist(sim[, , i]) |> mean()
  })

  (obs_dist - mean(sim_mean_dist)) / sd(sim_mean_dist)
}

diss <- split(arth, arth$site) |>
  sapply(FUN = function(dat) {
    summarize(dat, abund = n(), .by = c(tree, species_code)) |>
      pivot_wider(names_from = species_code,
                  values_from = abund,
                  values_fill = 0) |>
```

```

        select(!tree) |>
        z_diss()
    })

mete_by_diss <- mutate(arth_mete,
                      mean_diss = diss[site])

ggplot(mete_by_diss, aes(mean_diss, z2_sad, color = site)) +
  geom_jitter(width = 0.25) +
  this_ms_look(log_y = TRUE)

xy <- mete_by_diss |>
  summarize(z2_sad_mean = mean(log(z2_sad)),
            z2_sad_sd = sd(log(z2_sad)),
            n = n(),
            mean_diss = first(mean_diss),
            .by = site)

rma(yi = z2_sad_mean,
    vi = z2_sad_sd^2 / n,
    mods = ~ mean_diss,
    data = xy,
    method = "REML")

```